

Mamba vs Transformer Long-Sequence Benchmark Results

Seq Len	Model	Test Acc	Test F1	Best Val F1	Epochs	Params	Train Time	Infer Time	Throughput	VRAM
256	Mamba	0.715	0.715	0.741	10	9.57M	16.77s	0.29s	371,789	153MB
256	PatchTST	0.633	0.633	0.648	8	12.15M	13.79s	0.26s	409,026	144MB
256	DeBERTa	0.695	0.693	0.769	10	141.31M	205.24s	1.90s	56,135	1715MB
512	Mamba	0.703	0.701	0.756	10	9.57M	24.55s	0.45s	317,678	115MB
512	PatchTST	0.664	0.651	0.704	7	12.15M	22.01s	0.46s	311,261	145MB
512	DeBERTa	0.730	0.730	0.783	10	141.31M	530.07s	5.13s	27,971	1902MB
1024	Mamba	0.730	0.726	0.775	6	9.57M	31.38s	0.78s	207,272	115MB
1024	PatchTST	0.676	0.675	0.702	10	12.15M	59.88s	0.69s	232,569	145MB
1024	DeBERTa	0.727	0.721	0.795	10	141.31M	1640.73s	17.00s	9,487	2424MB
2048	Mamba	0.757	0.757	-	-	9.57M	94s	1.37s	118,000	151MB
2048	PatchTST	0.674	0.674	-	-	12.15M	116s	1.21s	135,000	145MB
2048	DeBERTa	OOM	OOM	OOM	OOM	141.31M	OOM	OOM	OOM	OOM

Key Insights

- 1. Mamba scales effectively as sequence length increases, with F1 improving from 0.715 at 256 tokens to 0.757 at 2048 tokens.
- 2. DeBERTa achieves strong performance at shorter sequences but suffers from severe computational and memory growth.
- 3. DeBERTa encounters Out-of-Memory (OOM) failure at sequence length 2048.
- 4. Mamba maintains competitive accuracy while using dramatically lower VRAM and training time.
- 5. PatchTST provides efficient throughput but weaker semantic performance for NLP classification.